B. Split

time limit per test: 1 second

memory limit per test: 256 megabytes

You are given a sequence a containing $2n$ integers. Let $f(b)$ denote the number of distinct elements with an odd number of occurrences in sequence b. You need to split the given array into two disjoint subsequences p and q, each of size n, such that $f(p) + f(q)$ is maximized. Output the **maximum** value.

A sequence a is a subsequence of a sequence b if a can be obtained from b by the deletion of several (possibly, zero or all) element from arbitrary positions.

Input

Each test contains multiple test cases. The first line contains the number of test cases t ($1 \le t \le 10^4$). The description of the test cases follows.

The first line of each test case contains an integer n ($1 \le n \le 2 \cdot 10^5$).

The second line contains $2n$ integers $a_1, a_2, \ldots, a_{2n}$ ($1 \le a_i \le 2n$) — the elements of sequence a.

It is guaranteed that the sum of n over all test cases does not exceed $2 \cdot 10^5$.

Output

For each test case, print one line.

You have to print the maximum value of $f(p) + f(q)$ that can be achieved.

Example

input Copy

```
7
2
1 2 3 4
3
5 5 5 5 5 5
4
3 3 7 6 3 7 8 7
2
2 2 2 2
6
1 2 3 4 5 4 1 4 1 5 4 6
4
1 2 1 2 1 2 1 2
5
9 9 9 7 7 7 9 7 7 7
```

output Copy

```
4
2
4
0
8
4
2
```

Note

For the first test case:

- We can divide the array such that $p = [1, 3]$ and $q = [2, 4]$.

→ Submit?

Language: GNU G++17 7.3.0

Choose file: Choose File No file chosen

Be careful: there is 50 points penalty for submission which fails the pretests or resubmission (except failure on the first test, denial of judgement or similar verdicts). "Passed pretests" submission verdict doesn't guarantee that the solution is absolutely correct and it will pass system tests.

Submit

→ Score table

	Score
Problem A	440
Problem B	1100
Problem C	1320
Problem D	1760
Problem E	2420
Problem F1	2200
Problem F2	880
Successful hack	100
Unsuccessful hack	-50
Unsuccessful submission	-50
Resubmission	-50

* If you solve problem on 00:30 from the first attempt

- This way, $f(p) = 2$ and $f(q) = 2$ since both have two distinct elements with odd frequency.

For the second test case:

- We can divide the array such that $p = [5, 5, 5]$ and $q = [5, 5, 5]$.
- This way, $f(p) = 1$ and $f(q) = 1$.

For the fifth test case:

- We can divide the array such that $p = [1, 2, 3, 4, 5, 6]$ and $q = [4, 1, 4, 1, 5, 4]$.
- This way, $f(p) = 6$ and $f(q) = 2$.
